

AI-Based Fire Fighting and Human Detection Rescue Robot Using Raspberry Pi and Arduino

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Abstract

Fire accidents and emergency situations often put human lives at risk. Entering hazardous environments during a fire can be extremely dangerous for firefighters and rescue teams. To address this problem, we developed an AI-Based Fire Fighting and Human Detection Rescue Robot using Arduino Uno and Raspberry Pi.

The robot can be controlled remotely through Bluetooth communication. A flame sensor is used to detect fire, while an ultrasonic sensor detects nearby obstacles. When fire is detected, a relay module automatically activates a water pump to extinguish the fire. Raspberry Pi performs human detection using computer vision techniques and sends alerts to the robot controller.

The proposed system is low-cost, efficient, and suitable for industrial safety, rescue operations, and emergency response applications.

1. Introduction

Fire accidents frequently occur in homes, industries, warehouses, and commercial buildings. During rescue operations, firefighters often face life-threatening situations while entering affected areas.

Recent developments in robotics and artificial intelligence have made it possible to create intelligent systems capable of assisting rescue teams. A rescue robot can reduce human risk by entering dangerous locations and performing essential tasks such as fire detection, obstacle monitoring, and human identification.

This project presents an AI-Based Fire Fighting and Human Detection Rescue Robot that combines Arduino Uno and Raspberry Pi to provide intelligent monitoring and automated fire-fighting capabilities.

2. Problem Statement

During fire emergencies, rescue personnel are exposed to dangerous conditions while attempting to save lives and property. Manual fire-fighting operations are risky, time-consuming, and often inefficient in hazardous environments.

Most low-cost fire-fighting robots available today lack advanced features such as human detection and intelligent monitoring. Therefore, there is a need for a smart robotic system capable of detecting fire, locating humans, and assisting rescue operations safely and effectively.

3. Objectives

The main objectives of this project are:

1. To develop a Bluetooth-controlled rescue robot.
 2. To detect fire using a flame sensor.
 3. To extinguish fire automatically using a water pump.
 4. To detect obstacles using an ultrasonic sensor.
 5. To detect humans using Raspberry Pi and AI-based computer vision.
 6. To improve safety and efficiency during rescue operations.
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4. Proposed System

The proposed system consists of two major units:

Arduino Unit

The Arduino Uno is responsible for:

- Bluetooth communication
- Motor control
- Flame detection

- Obstacle detection
- Relay control
- Water pump activation
- Buzzer control

Raspberry Pi Unit

The Raspberry Pi is responsible for:

- Camera processing
- Human detection
- AI-based image analysis
- Communication with Arduino

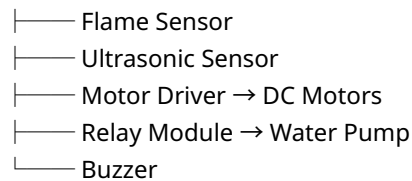
The combined system allows intelligent monitoring and automated emergency response.

5. Block Diagram

Bluetooth App



Arduino Uno



Raspberry Pi



Camera Module



Human Detection System



Arduino Uno

6. Hardware Components

Component	Quantity
Arduino Uno R3	1
Raspberry Pi	1
HC-05 Bluetooth Module	1

Component	Quantity
Flame Sensor	1
Ultrasonic Sensor HC-SR04	1
BTS7960 Motor Driver	2
DC Motors	2
Relay Module	1
Water Pump	1
Active Buzzer	1
Li-ion Battery	1
Jumper Wires	Multiple

7. Software Tools

The following software tools were used during development:

- Arduino IDE
- Raspberry Pi OS
- Python Programming Language
- OpenCV Library
- YOLO Human Detection Model
- Bluetooth RC Controller Application

8. Circuit Connections

HC-05 Bluetooth Module

TX → Arduino D2
 RX → Arduino D3
 VCC → 5V
 GND → GND

Flame Sensor

OUT → D9
 VCC → 5V
 GND → GND

Ultrasonic Sensor

TRIG → A4
ECHO → A5
VCC → 5V
GND → GND

Relay Module

IN → D4
VCC → 5V
GND → GND

Left Motor Driver

RPWM → A0
LPWM → A1
REN → A2
LEN → A3

Right Motor Driver

RPWM → D5
LPWM → D6
REN → D7
LEN → D8

9. Working Principle

Bluetooth Control

The robot receives commands from a smartphone via Bluetooth communication.

Commands:

- F = Forward
- B = Backward
- L = Left
- R = Right
- S = Stop

Fire Detection

The flame sensor continuously monitors the environment.

When fire is detected:

- Buzzer activates
- Relay turns ON
- Water pump starts automatically
- Fire extinguishing process begins

Obstacle Detection

The ultrasonic sensor continuously measures distance.

When an obstacle is detected:

- Warning buzzer activates
- Robot stops or alerts the operator

Human Detection

The Raspberry Pi camera continuously captures images.

Using AI-based computer vision:

- Human presence is detected
- Alert messages are generated
- Robot can stop automatically for safety

10. Results and Testing

Several tests were conducted to verify system performance.

Test 1: Bluetooth Control

The robot successfully responded to all movement commands sent from the mobile application.

Test 2: Fire Detection

The flame sensor successfully detected fire and activated the relay module and water pump.

Test 3: Obstacle Detection

The ultrasonic sensor successfully detected nearby obstacles and generated warning alerts.

Test 4: Human Detection

The Raspberry Pi successfully identified human presence using computer vision techniques.

Test 5: Water Pump Activation

The relay module successfully controlled the water pump during fire detection events.

11. Real-Life Applications

This project can be used in:

- Fire rescue operations
- Industrial safety monitoring
- Warehouses and factories
- Chemical plants
- Smart safety systems
- Emergency response operations

The system improves safety by reducing direct human exposure to dangerous environments.

12. Advantages

- Low-cost implementation
 - Easy operation
 - Wireless control
 - Automatic fire extinguishing
 - Human detection capability
 - Obstacle detection capability
 - Real-time monitoring
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13. Limitations

- Limited battery backup
 - Human detection performance decreases in low-light conditions
 - Suitable for small-scale fire incidents
 - Requires stable power supply
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14. Cost Analysis

Component	Cost (BDT)
Arduino Uno	700
Raspberry Pi	9000
Motors	800
Motor Drivers	1000
Sensors	700
Relay and Pump	600
Battery	1200
Other Components	1000

Total Cost $\approx$ 15,000 BDT

The project is cost-effective compared to commercial rescue robots while still providing multiple intelligent features.

15. Future Work

The following improvements can be implemented in future versions:

- GPS Tracking System
 - IoT-based Monitoring
 - Mobile Application Integration
 - Autonomous Navigation
 - Cloud Data Logging
 - Live Video Streaming
 - Advanced AI Object Detection
 - Real-Time Location Sharing
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16. Conclusion

This project successfully developed an AI-Based Fire Fighting and Human Detection Rescue Robot capable of detecting fire, obstacles, and human presence. The integration of Arduino Uno and Raspberry Pi provides intelligent monitoring, wireless control, and automated fire-fighting functionality.

The project demonstrates the practical application of Artificial Intelligence, Embedded Systems, and Robotics in improving safety and rescue operations. Future enhancements can further improve system autonomy, reliability, and real-world applicability.

References

1. Arduino Official Documentation
2. Raspberry Pi Official Documentation
3. OpenCV Documentation
4. YOLO Object Detection Documentation
5. HC-SR04 Ultrasonic Sensor Datasheet
6. HC-05 Bluetooth Module Datasheet
7. GitHub Repository Link
8. Project Demonstration Video Link